

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 05

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B - rivers are accessible over a large area with few natural contaminants (1)	1
01.2	<i>1 mark for each point.</i> • Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)	2
01.3	<i>1 mark for each point.</i> • Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
02.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks (1) • Kills most pathogens (1) • Reduces odour and volume of the sludge (1) • Produces biogas (mainly methane with some CO ₂) used as fuel to heat the digester / generate electricity (1) • Digested sludge has a better C:N ratio and can be used as agricultural fertiliser (1)	4

Question	Marking guidance	Marks
02.2	<i>1 mark for each point.</i> • Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility (1) • Remaining pathogens - destroyed by heat treatment of the sludge before use (1) • Timing: spreading in the growing season covers crops; in winter vehicles compact wet soils - apply at suitable times / when soils are dry (1) • Odour / public objection - digestion reduces odour (1)	4
02.3	<i>1 mark for each point.</i> • Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built; surface water needs little treatment (1) • The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	
03.1	<i>Correct answer only.</i> • B - Low liposolubility and low persistence (1)	1
03.2	<i>1 mark for each point.</i> • Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown (1) • Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain (1)	2
03.3	<i>1 mark for each point.</i> • They are intended to be toxic (to the target species) (1) • They are deliberately released / spread into the environment (rather than being an accidental or waste release) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
04.1	<i>1 mark for each point.</i> • Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes (1) • Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH (1) • Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow (1) • Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser (1) • Unattractive topography - landscaping / reshaping (1) • Dust / heavy metals - cover and keep at high pH (1)	6
04.2	<i>1 mark for each point.</i> • So that plant roots cannot reach the toxic layer (1) • Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains (1)	2
04.3	<i>1 mark for each point.</i> • Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions (1) • Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
05.1	<i>1 mark for each point.</i> - Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury (1) - Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes (1) - It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water (1) - Local people ate large amounts of fish (critical group) so received toxic doses (1)	4
05.2	<i>1 mark for each point.</i> - Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) (1) - It crosses the placenta and harms the developing unborn baby (1) - Mercury is a teratogen, preventing normal gene expression / development (1)	3
05.3	<i>1 mark for each point.</i> - Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs (1) - Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut (1) - Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs; liposoluble so most toxic and bioaccumulates (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
06.1	<i>1 mark for each point.</i> - Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated (1) - Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time (1) - Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it (1) - Repeat each treatment several times and calculate means (and standard deviation) (1) - Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance (1)	5
06.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Rainfall intensity / raindrop size / height (1) - Soil texture, compaction, organic matter, depth (1) - Permeability of the material beneath (1) - Gradient (1) - Root binding (1)	3
06.3	<i>1 mark for each point.</i> - All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead (1) - Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	
07.1	<i>Correct answer only.</i> • A waste is assumed to cause pollution until research shows it is unlikely to do so (1)	1
07.2	<i>1 mark for each point.</i> • If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution (1) • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply (1)	2
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marine discharges where currents dilute and disperse emissions (1) • Emissions downwind of urban areas (1) • Not discharging waste onto permeable rock above an aquifer (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
08.1	<i>1 mark for each point.</i> • Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium (1) • Positive feedback: a change causes other changes that increase the original change (1)	2
08.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced (2) • Increased photosynthesis: higher temperatures and CO ₂ increase photosynthesis, removing CO ₂ ; if stored in woody tissue atmospheric CO ₂ rises less (2)	4
08.3	<i>1 mark for each point.</i> • Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground (1) • Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming (1) • Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature (1) • If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming (1)	4
08.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Reduced albedo as snow / ice melts (1) • Ocean acidification reducing coral survival and carbon sequestration (1) • Increased forest and peat fires releasing CO ₂ (1) • Faster decomposition of soil organic matter releasing CO ₂ (1) • Increased water vapour (a greenhouse gas) in warmer air (1) • Increased cirrus (high) cloud which traps infrared (1)	3

Question	Marking guidance	Marks
08.5	<i>1 mark for each point.</i> • A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable (1) • So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
09.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises (2) • Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) (2) • Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises (2)	6
09.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Rice paddy fields (1) • Landfill sites (1) • Anaerobic sediments in reservoirs (1) • Livestock intestines (cattle) (1) • Manure / slurry stores (1)	3
09.3	<i>1 mark.</i> • More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
10.1	<i>1 mark for each point.</i> • Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) (1) • They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone (1) • The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface (1)	3
10.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Provides carbon, oxygen, hydrogen and nitrogen (as N₂, O₂, CO₂, H₂O) for biological molecules (1) • Delays the escape of infrared energy, keeping the surface warm (1) • Distributes heat and water vapour by winds / ocean currents (1) • Provides atmospheric pressure so liquid water can exist (1) • Deflects damaging radiation of the solar wind in the upper atmosphere (1)	2
10.3	<i>1 mark.</i> • Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers (1)	1

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO₂, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	

Question	Marking guidance	Marks
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO ₂ and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25